

12 System configuration controller (SYSCFG)

The STM32F3xx devices feature a set of configuration registers. The main purposes of the system configuration controller are the following:

- Enabling/disabling I²C Fm+ on some I/O ports
- Remapping some DMA trigger sources from TIM16, TIM17, TIM6, DAC1_CH1, and DAC1_CH2, TIM7, and ADC4 to different DMA channels (also SPI1, I2C1, ADC2 in STM32F303x6/8 and STM32F328x8)
- Remapping the memory located at the beginning of the code area
- Managing the external interrupt line connection to the GPIOs
- Remapping TIM1 ITR3 source
- Remapping USB interrupt line
- Remapping DAC1 and DAC2 triggers
- Managing robustness feature
- Configuring encoder mode
- CCM SRAM pages protection

12.1 SYSCFG registers

12.1.1 SYSCFG configuration register 1 (SYSCFG_CFGR1)

This register is used for specific configurations on memory remap.

Two bits are used to configure the type of memory accessible at address 0x0000 0000. These bits are used to select the physical remap by software and so, bypass the BOOT pin and the option bit setting.

After reset these bits take the value selected by the BOOT pin (BOOT0) and by the option bit (BOOT1).

Address offset: 0x00

Reset value: 0x7C00 000X (X is the memory mode selected by the BOOT0 pin and BOOT1 option bit)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FPU_IE[5..0]						Res	I2C3_FMP	ENCODER_MODE	I2C2_FMP	I2C1_FMP	I2C_PB9_FMP	I2C_PB8_FMP	I2C_PB7_FMP	I2C_PB6_FMP	
rw	rw	rw	rw	rw	rw			rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DAC2_CH1_DMA_RMP ⁽¹⁾	TIM7_DAC2_DMA_RMP	TIM6_DAC1_DMA_RMP	TIM17_DMA_RMP	TIM16_DMA_RMP	Res	Res	ADC2_DMA_RMP	DAC_TRIG_RMP	TIM1_ITR3_RMP	USB_IT_RMP	Res	Res	MEM_MODE ⁽²⁾	MEM_MODE	
rw	rw	rw	rw	rw			rw	rw	rw	rw				rw	rw

1. Only in STM32F303x6/8 and STM32F328x8.

2. Only for STM32F303xD/E and STM32F398xE devices

- Bits 31:26 **FPU_IE[5..0]**: Floating Point Unit interrupts enable bits
 FPU_IE[5]: Inexact interrupt enable
 FPU_IE[4]: Input normal interrupt enable
 FPU_IE[3]: Overflow interrupt enable
 FPU_IE[2]: underflow interrupt enable
 FPU_IE[1]: Divide-by-zero interrupt enable
 FPU_IE[0]: Invalid operation interrupt enable
- Bit 25: Reserved, must be kept at reset value.
- Bit 24 **I2C3_FMP**: I2C3 fast mode Plus driving capability activation (STM32F303xD/E devices only)
 This bit is set and cleared by software. It enables the Fm+ on I2C3 pins selected through AF selection bits.
 0: Fm+ mode is not enabled on I2C3 pins selected through AF selection bits
 1: Fm+ mode is enabled on I2C3 pins selected through AF selection bits.
- Bits 23:22 **ENCODER_MODE**: Encoder mode
 This bit is set and cleared by software.
 00: No redirection.
 01: TIM2 IC1 and TIM2 IC2 are connected to TIM15 IC1 and TIM15 IC2 respectively.
 10: TIM3 IC1 and TIM3 IC2 are connected to TIM15 IC1 and TIM15 IC2 respectively.
 11: TIM4 IC1 and TIM4 IC2 are connected to TIM15 IC1 and TIM15 IC2 respectively (STM32F303xB/C and STM32F358xC devices only).
- Bit 21 **I2C2_FMP**: I2C2 fast mode Plus driving capability activation (STM32F303xB/C and STM32F358xC devices only)
 This bit is set and cleared by software. It enables the Fm+ on I2C2 pins selected through AF selection bits.
 0: Fm+ mode is not enabled on I2C2 pins selected through AF selection bits
 1: Fm+ mode is enabled on I2C2 pins selected through AF selection bits.
- Bit 20 **I2C1_FMP**: I2C1 Fm+ driving capability activation
 This bit is set and cleared by software. It enables the Fm+ on I2C1 pins selected through AF selection bits.
 0: Fm+ mode is not enabled on I2C1 pins selected through AF selection bits
 1: Fm+ mode is enabled on I2C1 pins selected through AF selection bits.
- Bits 19:16 **I2C_PbX_FMP**: Fm+ driving capability activation on the pad
 These bits are set and cleared by software. Each bit enables I²C Fm+ mode for PB6, PB7, PB8, and PB9 I/Os.
 0: PBx pin operates in standard mode (Sm), x = 6..9
 1: I²C Fm+ mode enabled on PBx pin, and the Speed control is bypassed.
- Bit 15 **DAC2_CH1_DMA_RMP**: DAC2 channel1 DMA remap (STM32F303x6/8 and STM32F328x8 devices only)
 This bit is set and cleared by software. It controls the remapping of DAC2 channel1 DMA request.
 0: No remap
 1: Remap (DAC2_CH1 DMA requests mapped on DMA1 channel 5)
Note: In STM32F303x6/8 and STM32F328x8, this bit must be set.

Bit 14 **TIM7_DAC1_CH2_DMA_RMP**: TIM7 and DAC channel2 DMA remap

This bit is set and cleared by software. It controls the remapping of TIM7(UP) and DAC channel2 DMA request.

0: No remap (TIM7_UP and DAC_CH2 DMA requests mapped on DMA2 channel 4 in STM32F303xB/C and STM32F358xC devices)

1: Remap (TIM7_UP and DAC_CH2 DMA requests mapped on DMA1 channel 4)

Note: In STM32F303x6/8 and STM32F328x8, this bit must be set as there is no DMA2 in these products.

Bits 15:14 Reserved, must be kept at reset value.

Bit 13 **TIM6_DAC1_CH1_DMA_RMP**: TIM6 and DAC channel1 DMA remap

This bit is set and cleared by software. It controls the remapping of TIM6 (UP) and DAC channel1 DMA request.

0: No remap (TIM6_UP and DAC_CH1 DMA requests mapped on DMA2 channel 3 in STM32F303xB/C and STM32F358xC)

1: Remap (TIM6_UP and DAC_CH1 DMA requests mapped on DMA1 channel 3)

Note: In STM32F303x6/8 and STM32F328x8, this bit must be set as there is no DMA2 in these products.

Bit 12 **TIM17_DMA_RMP**: TIM17 DMA request remapping bit

This bit is set and cleared by software. It controls the remapping of TIM17 DMA request.

0: No remap (TIM17_CH1 and TIM17_UP DMA requests mapped on DMA1 channel 1)

1: Remap (TIM17_CH1 and TIM17_UP DMA requests mapped on DMA1 channel 7)

Bit 11 **TIM16_DMA_RMP**: TIM16 DMA request remapping bit

This bit is set and cleared by software. It controls the remapping of TIM16 DMA request.

0: No remap (TIM16_CH1 and TIM16_UP DMA requests mapped on DMA1 channel 3)

1: Remap (TIM16_CH1 and TIM16_UP DMA requests mapped on DMA1 channel 6)

Bits 10:9 Reserved, must be kept at reset value.

Bit 8 **ADC2_DMA_RMP**: ADC2 DMA remapping bit

This bit is set and cleared by software. It controls the remapping of ADC24 DMA requests.

0: No remap (ADC24 DMA requests mapped on DMA2 channels 1 and 2)

1: Remap (ADC24 DMA requests mapped on DMA2 channels 3 and 4)

Bit 7 **DAC1_TRIG_RMP**: DAC trigger remap (when TSEL = 001) This bit is set and cleared by software. It controls the mapping of the DAC trigger source.

0: No remap (DAC trigger is TIM8_TRGO in STM32F303xB/C and STM32F358xC devices)

1: Remap (DAC trigger is TIM3_TRGO)

Bit 6 **TIM1_ITR3_RMP**: Timer 1 ITR3 selection

This bit is set and cleared by software. It controls the mapping of TIM1 ITR3.

0: No remap (TIM1_ITR3 = TIM4_TRGO in STM32F303xB/C and STM32F358xC devices)

1: Remap (TIM1_ITR3 = TIM17_OC)

Bit 5 **USB_IT_RMP**: USB interrupt remap (STM32F303xB/C/D/E devices only)

This bit is set and cleared by software. It controls the USB interrupts mapping.

0: USB_HP, USB_LP and USB_WAKEUP interrupts are mapped on interrupt lines 19, 20 and 42 respectively.

1: USB_HP, USB_LP and USB_WAKEUP interrupts are mapped on interrupt lines 74, 75 and 76 respectively.

Bits 4:3 Reserved, must be kept at reset value.

Bits 2:0 **MEM_MODE**: Memory mapping selection bits

This bit is set and cleared by software. It controls the memory internal mapping at address 0x0000 0000. After reset these bits take on the memory mapping selected by BOOT0 pin and BOOT1 option bit.

0x0: Main Flash memory mapped at 0x0000 0000

001: System Flash memory mapped at 0x0000 0000

011: Embedded SRAM (on the D-Code bus) mapped at 0x0000 0000

1xx: FMC Bank (Only the first two banks) (Available on STM32F303xD/E only).

12.1.2 SYSCFG CCM SRAM protection register (SYSCFG_RCR)

The CCM SRAM has a size of 8 Kbytes, organized in 8 pages (1 Kbyte each) in STM32F303xB/C and STM32F358xC devices. The CCM SRAM has a size of 4 Kbytes, organized in 4 pages (1 Kbyte each) in STM32F303x6/8 and STM32F328x8. The CCM SRAM has a size of 16 Kbytes, organized in 16 pages (1 Kbyte each) in STM32F303xD/E.

Each page can be write protected.

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PAGE15 _WP ⁽¹⁾	PAGE14 _WP ⁽¹⁾	PAGE13 _WP ⁽¹⁾	PAGE12 _WP ⁽¹⁾	PAGE11 _WP ⁽¹⁾	PAGE10 _WP ⁽¹⁾	PAGE9 _WP ⁽¹⁾	PAGE8 _WP ⁽¹⁾	PAGE7 _WP ⁽²⁾	PAGE6 _WP ⁽²⁾	PAGE5 _WP ⁽²⁾	PAGE4 _WP ⁽²⁾	PAGE3 _WP	PAGE2 _WP	PAGE1 _WP	PAGE0 _WP
												rw	rw	rw	rw

1. Only on STM32F303xD/E and STM32F398xE

2. Only on STM32F303xB/C/D/E devices.

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **PAGEx_WP** (x= 0 to 15): CCM SRAM page write protection bit)

These bits are set by software. They can be cleared only by system reset.

0: Write protection of pagex is disabled.

1: Write protection of pagex is enabled.

12.1.3 SYSCFG external interrupt configuration register 1 (SYSCFG_EXTICR1)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI3[3:0]				EXTI2[3:0]				EXTI1[3:0]				EXTI0[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw



Bits 31:16 Reserved, must be kept at reset value.

Bits 15:12 **EXTI3[3:0]**: EXTI 3 configuration bits

These bits are written by software to select the source input for the EXTI3 external interrupt.

x000: PA[3] pin

x001: PB[3] pin

x010: PC[3] pin

x011: PD[3] pin

x100: PE[3] pin

x101: PF[3] pin

x110: PG[3] pin

other configurations: reserved

Bits 11:8 **EXTI2[3:0]**: EXTI 2 configuration bits

These bits are written by software to select the source input for the EXTI2 external interrupt.

x000: PA[2] pin

x001: PB[2] pin

x010: PC[2] pin

x011: PD[2] pin

x100: PE[2] pin

x101: PF[2] pin

x110: PG[2] pin

x111: PH[2] pin

other configurations: reserved

Bits 7:4 **EXTI1[3:0]**: EXTI 1 configuration bits

These bits are written by software to select the source input for the EXTI1 external interrupt.

x000: PA[1] pin

x001: PB[1] pin

x010: PC[1] pin

x011: PD[1] pin

x100: PE[1] pin

x101: PF[1] pin

x110: PG[1] pin

x111: PH[1] pin

other configurations: reserved

Bits 3:0 **EXTI0[3:0]**: EXTI 0 configuration bits

These bits are written by software to select the source input for the EXTI0 external interrupt.

x000: PA[0] pin

x001: PB[0] pin

x010: PC[0] pin

x011: PD[0] pin

x100: PE[0] pin

x101: PF[0] pin

x110: PG[0] pin

x111: PH[0] pin

Note: other configurations: reserved

12.1.4 SYSCFG external interrupt configuration register 2 (SYSCFG_EXTICR2)

Address offset: 0x0C
Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI7[3:0]				EXTI6[3:0]				EXTI5[3:0]				EXTI4[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw



Bits 31:16 Reserved, must be kept at reset value.

Bits 15:12 **EXTI7[3:0]**: EXTI 7 configuration bits

These bits are written by software to select the source input for the EXTI7 external interrupt.

x000: PA[7] pin

x001: PB[7] pin

x010: PC[7] pin

x011: PD[7] pin

x100: PE[7] pin

x101: PF[7] pin

x110: PG[7] pin

Other configurations: reserved

Bits 11:8 **EXTI6[3:0]**: EXTI 6 configuration bits

These bits are written by software to select the source input for the EXTI6 external interrupt.

x000: PA[6] pin

x001: PB[6] pin

x010: PC[6] pin

x011: PD[6] pin

x100: PE[6] pin

x101: PF[6] pin

x110: PG[6] pin

Other configurations: reserved

Bits 7:4 **EXTI5[3:0]**: EXTI 5 configuration bits

These bits are written by software to select the source input for the EXTI5 external interrupt.

x000: PA[5] pin

x001: PB[5] pin

x010: PC[5] pin

x011: PD[5] pin

x100: PE[5] pin

x101: PF[5] pin

x110: PG[5] pin

Other configurations: reserved

Bits 3:0 **EXTI4[3:0]**: EXTI 4 configuration bits

These bits are written by software to select the source input for the EXTI4 external interrupt.

x000: PA[4] pin

x001: PB[4] pin

x010: PC[4] pin

x011: PD[4] pin

x100: PE[4] pin

x101: PF[4] pin

x110: PG[4] pin

Other configurations: reserved

Note: Some of the I/O pins mentioned in the above register may not be available on small packages.

12.1.5 SYSCFG external interrupt configuration register 3 (SYSCFG_EXTICR3)

Address offset: 0x10

Reset value: 0x0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI11[3:0]				EXTI10[3:0]				EXTI9[3:0]				EXTI8[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:12 **EXTI11[3:0]**: EXTI 11 configuration bits

These bits are written by software to select the source input for the EXTI11 external interrupt.

x000: PA[11] pin
 x001: PB[11] pin
 x010: PC[11] pin
 x011: PD[11] pin
 x100: PE[11] pin
 x101: PF[11] pin
 x110: PG[11] pin
 other configurations: reserved

Bits 11:8 **EXTI10[3:0]**: EXTI 10 configuration bits

These bits are written by software to select the source input for the EXTI10 external interrupt.

x000: PA[10] pin
 x001: PB[10] pin
 x010: PC[10] pin
 x011: PD[10] pin
 x100: PE[10] pin
 x101: PF[10] pin
 x110: PG[10] pin
 other configurations: reserved

Bits 7:4 **EXTI9[3:0]**: EXTI 9 configuration bits

These bits are written by software to select the source input for the EXTI9 external interrupt.

x000: PA[9] pin
 x001: PB[9] pin
 x010: PC[9] pin
 x011: PD[9] pin
 x100: PE[9] pin
 x101: PF[9] pin
 x110: PG[9] pin
 other configurations: reserved

Bits 3:0 **EXTI8[3:0]**: EXTI 8 configuration bits

These bits are written by software to select the source input for the EXTI8 external interrupt.

x000: PA[8] pin
 x001: PB[8] pin
 x010: PC[8] pin
 x011: PD[8] pin
 x100: PE[8] pin
 x101: PF[8] pin
 x110: PG[8] pin
 other configurations: reserved

Note: Some of the I/O pins mentioned in the above register may not be available on small packages.

12.1.6 SYSCFG external interrupt configuration register 4 (SYSCFG_EXTICR4)

Address offset: 0x14

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI15[3:0]				EXTI14[3:0]				EXTI13[3:0]				EXTI12[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:12 **EXTI15[3:0]**: EXTI15 configuration bits

These bits are written by software to select the source input for the EXTI15 external interrupt.

x000: PA[15] pin
 x001: PB[15] pin
 x010: PC[15] pin
 x011: PD[15] pin
 x100: PE[15] pin
 x101: PF[15] pin
 x110: PG[15] pin
 Other configurations: reserved

Bits 11:8 **EXTI14[3:0]**: EXTI14 configuration bits

These bits are written by software to select the source input for the EXTI14 external interrupt.

x000: PA[14] pin
 x001: PB[14] pin
 x010: PC[14] pin
 x011: PD[14] pin
 x100: PE[14] pin
 x101: PF[14] pin
 x110: PG[14] pin

Other configurations: reserved

Bits 7:4 **EXTI13[3:0]**: EXTI13 configuration bits

These bits are written by software to select the source input for the EXTI13 external interrupt.

x000: PA[13] pin
 x001: PB[13] pin
 x010: PC[13] pin
 x011: PD[13] pin
 x100: PE[13] pin
 x101: PF[13] pin
 x110: PG[13] pin

Other configurations: reserved

Bits 3:0 **EXTI12[3:0]**: EXTI12 configuration bits

These bits are written by software to select the source input for the EXTI12 external interrupt.

x000: PA[12] pin
 x001: PB[12] pin
 x010: PC[12] pin
 x011: PD[12] pin
 x100: PE[12] pin
 x101: PF[12] pin
 x110: PG[12] pin

Other configurations: reserved

Note: Some of the I/O pins mentioned in the above register may not be available on small packages.

12.1.7 SYSCFG configuration register 2 (SYSCFG_CFGR2)

Address offset: 0x18

System reset value: 0x0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res	Res	Res	Res	Res	Res	Res	SRAM_PEF	Res	Res	Res	BYP_ADDR_PAR	Res	PVD_LOCK	SRAM_PARITY_LOCK	LOCKUP_LOCK
							rc_w1				rw		rw	rw	rw

Bits 31:9 Reserved, must be kept at reset value

Bit 8 **SRAM_PEF**: SRAM parity error flag

This bit is set by hardware when an SRAM parity error is detected. It is cleared by software by writing '1'.

0: No SRAM parity error detected

1: SRAM parity error detected

Bits 7:5 Reserved, must be kept at reset value

Bit 4 **BYP_ADDR_PAR**: Bypass address bit 29 in parity calculation

This bit is set by software and cleared by a system reset. It is used to prevent an unwanted parity error when the user writes a code in the RAM at address 0x2XXXXXXX (address in the address range 0x20000000-0x20002000) and then executes the code from RAM at boot (RAM is remapped at address 0x00). In this case, a read operation is performed from the range 0x00000000-0x00002000 resulting in a parity error (the parity on the address is different).

0: The ramload operation is performed taking into consideration bit 29 of the address when the parity is calculated.

1: The ramload operation is performed without taking into consideration bit 29 of the address when the parity is calculated.

Bit 3 Reserved, must be kept at reset value

Bit 2 **PVD_LOCK**: PVD lock enable bit

This bit is set by software and cleared by a system reset. It can be used to enable and lock the PVD connection to TIM1/8/15/16/17 Break input, as well as the PVDE and PLS[2:0] in the PWR_CR register.

0: PVD interrupt disconnected from TIM1/8/15/16/17 Break input. PVDE and PLS[2:0] bits can be programmed by the application.

1: PVD interrupt connected to TIM1/8/15/16/17 Break input, PVDE and PLS[2:0] bits are read only.

Bit 1 **SRAM_PARITY_LOCK**: SRAM parity lock bit

This bit is set by software and cleared by a system reset. It can be used to enable and lock the SRAM parity error signal connection to TIM1/8/15/16/17 Break inputs.

0: SRAM parity error signal disconnected from TIM1/8/15/16/17 Break inputs

1: SRAM parity error signal connected to TIM1/8/15/16/17 Break inputs

Bit 0 **LOCKUP_LOCK**: Cortex®-M4 LOCKUP (Hardfault) output enable bit

This bit is set by software and cleared by a system reset. It can be used to enable and lock the connection of Cortex®-M4 LOCKUP (Hardfault) output to TIM1/15/16/17 Break input.

0: Cortex®-M4 LOCKUP output disconnected from TIM1/8/15/16/17 Break inputs.

1: Cortex®-M4 LOCKUP output connected to TIM1/8/15/16/17 Break inputs

12.1.8 SYSCFG configuration register 3 (SYSCFG_CFGR3)

Note: This register is available in STM32F303x6/x8 and STM32F328 devices only.

Address offset: 0x50

System reset value: 0x0000 0200

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res	Res	Res	Res	Res	Res	ADC2_DMA_RMP	I2C1_TX_DMA_RMP	I2C1_RX_DMA_RMP	SPI1_TX_DMA_RMP	SPI1_RX_DMA_RMP					

Bits 31:10 Reserved, must be kept at reset value

Bit 9 **ADC2_DMA_RMP[1]**: ADC2 DMA controller remapping bit

0: ADC2 mapped on DMA2

1: ADC2 mapped on DMA1

Bit 8 **ADC2_DMA_RMP[0]**: ADC2 DMA channel remapping bit

0: ADC2 mapped on DMA1 channel 2

1: ADC2 mapped on DMA1 channel 4

Bits 7:6 **I2C1_TX_DMA_RMP**: I2C1_TX DMA remapping bit

This bit is set and cleared by software. It defines on which DMA1 channel I2C1_TX is mapped.

00: I2C1_TX mapped on DMA1 CH6

01: I2C1_TX mapped on DMA1 CH2

10: I2C1_TX mapped on DMA1 CH4

11: I2C1_TX mapped on DMA1 CH6

Bits 5:4 **I2C1_RX_DMA_RMP**: I2C1_RX DMA remapping bit

This bit is set and cleared by software. It defines on which DMA1 channel I2C1_RX is mapped.

00: I2C1_RX mapped on DMA1 CH7

01: I2C1_RX mapped on DMA1 CH3

10: I2C1_RX mapped on DMA1 CH5

11: I2C1_RX mapped on DMA1 CH7

Bits 3:2 **SPI1_TX_DMA_RMP**: SPI1_TX DMA remapping bit

This bit is set and cleared by software. It defines on which DMA1 channel SPI1_TX is mapped.

00: SPI1_TX mapped on DMA1 CH3

01: SPI1_TX mapped on DMA1 CH5

10: SPI1_TX mapped on DMA1 CH7

11: SPI1_TX mapped on DMA1 CH3

Bits 1:0 **SPI1_RX_DMA_RMP**: SPI1_RX DMA remapping bit

This bit is set and cleared by software. It defines on which DMA1 channel SPI1_RX is mapped.

00: SPI1_RX mapped on DMA1 CH2

01: SPI1_RX mapped on DMA1 CH4

10: SPI1_RX mapped on DMA1 CH6

11: SPI1_RX mapped on DMA1 CH2

12.1.9 SYSCFG configuration register 4 (SYSCFG_CFGR4)

Note: This register is available in STM32F303xD/E and STM32F398xE devices only.

SYSCFG_CFGR4 is added allowing to remap the triggers of the ADCs, mainly the new TIM20 events.

Address offset: 0x48

System reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res	Res	ADC34_JEXT14_RMP	ADC34_JEXT11_RMP	ADC34_JEXT5_RMP	ADC34_EXT15_RMP	ADC34_EXT6_RMP	ADC34_EXT5_RMP	ADC12_JEXT13_RMP	ADC12_JEXT6_RMP	ADC12_JEXT3_RMP	ADC12_EXT15_RMP	ADC12_EXT13_RMP	ADC12_EXT5_RMP	ADC12_EXT3_RMP	ADC12_EXT2_RMP
		r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:14 Reserved, must be kept at reset value

Bit 13 **ADC34_JEXT14_RMP** - Controls the Input trigger of ADC34 injected channel JEXT14:

- 0: Trigger source is TIM7_TRGO
- 1: Trigger source is TIM20_CC2

Bit 12 **ADC34_JEXT11_RMP** - Controls the Input trigger of ADC34 injected channel JEXT11:

- 0: Trigger source is TIM1_CC3
- 1: Trigger source is TIM20_TRGO2

Bit 11 **ADC34_JEXT5_RMP** - Controls the Input trigger of ADC34 injected channel JEXT5:

- 0: Trigger source is TIM4_CC3
- 1: Trigger source is TIM20_TRGO

Bit 10 **ADC34_EXT15_RMP** - Controls the Input trigger of ADC34 regular channel EXT15:

- 0: Trigger source is TIM2_CC1
- 1: Trigger source is TIM20_CC1

Bit 9 **ADC34_EXT6_RMP** - Controls the Input trigger of ADC34 regular channel EXT6:

- 0: Trigger source is TIM4_CC1
- 1: Trigger source is TIM20_TRGO2

Bit 8 **ADC34_EXT5_RMP** - Controls the Input trigger of ADC34 regular channel EXT5:

- 0: Trigger source is EXTI line 2 when reset at '0'
- 1: Trigger source is TIM20_TRGO

Bit 7 **ADC12_JEXT13_RMP** - Controls the Input trigger of ADC12 injected channel JEXT13:

- 0: Trigger source is TIM3_CC1
- 1: Trigger source is TIM20_CC4

Bit 6 **ADC12_JEXT6_RMP** - Controls the Input trigger of ADC12 injected channel JEXT6:

- 0: Trigger source is EXTI line 15
- 1: Trigger source is TIM20_TRGO2

Bit 5 **ADC12_JEXT3_RMP** - Controls the Input trigger of ADC12 injected channel EXT3:

- 0: Trigger source is TIM2_CC1
- 1: Trigger source is TIM20_TRGO

Bit 4 **ADC12_EXT15_RMP** - Controls the Input trigger of ADC12 regular channel EXT15:

- 0: Trigger source is TIM3_CC4
- 1: Trigger source is TIM20_CC3

Bit 3 **ADC12_EXT13_RMP** - Controls the Input trigger of ADC12 regular channel EXT13:

- 0: Trigger source is TIM6_TRGO
- 1: Trigger source is TIM20_CC2

Bit 2 **ADC12_EXT5_RMP** - Controls the Input trigger of ADC12 regular channel EXT5

- 0: Trigger source is TIM4_CC4
- 1: Trigger source is TIM20_CC1

Bit 1 **ADC12_EXT3_RMP** - Controls the Input trigger of ADC12 regular channel EXT3:

- 0: Trigger source is TIM2_CC2
- 1: Trigger source is TIM20_TRGO2

Bit 0 **ADC12_EXT2_RMP** - Controls the Input trigger of ADC12 regular channel EXT2:

- 0: Trigger source is TIM1_CC3
- 1: Trigger source is TIM20_TRGO

12.1.10 SYSCFG register map

Table 74. SYSCFG register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	SYSCFG_CFGR1	FPU_IE[5..0]						Res	Res	ENCODER_MODE [1:0]		I2C2_FMP	I2C1_FMP	I2C_PB9_FMP	I2C_PB8_FMP	I2C_PB7_FMP	I2C_PB6_FMP	DAC2_CH1_DMA_RMP	TIM7_DAC2_DMA_RMP	TIM6_DAC1_DMA_RMP	TIM17_DMA_RMP	TIM16_DMA_RMP	Res	Res	ADC24_DMA_RMP	DAC_TRIG_RMP	TIM1_ITR3_RMP	USB_IT_RMP	Res	Res	Res	MEM_MODE	
	Reset value	1	1	1	1	1	0			0	0	0	0	0	0	0	0	0	0	0	0	0			0	0	0	0				X	X
0x04	SYSCFG_RCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PAGE[15:0]_WP															
	Reset value																		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x08	SYSCFG_EXTICR1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	EXTI3[3:0]			EXTI2[3:0]			EXTI1[3:0]			EXTI0[3:0]						
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0C	SYSCFG_EXTICR2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	EXTI7[3:0]			EXTI6[3:0]			EXTI5[3:0]			EXTI4[3:0]						
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 74. SYSCFG register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0x10	SYSCFG_EXTICR3	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	EXTI11[3:0]				EXTI10[3:0]				EXTI9[3:0]				EXTI8[3:0]					
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x14	SYSCFG_EXTICR4	Res.	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	EXTI15[3:0]				EXTI14[3:0]				EXTI13[3:0]				EXTI12[3:0]					
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x18	SYSCFG_CFGR2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	SRAM_PEF				Res	Res	Res	BYP_ADDR_PAR	Res	PVD LOCK	SRAM_PARITY_LOCK	LOCKUP_LOCK
	Reset value																							0			0		0		0	0	0	0	
.	.																																		
0x50	SYSCFG_CFGR3	Res.	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res.	Res.	Res.	Res.	Res	Res	Res	ADC2_DMA_RMP				I2C1_TX_DMA_RMP				I2C1_RX_DMA_RMP				
	Reset value																							1	0	0	0	0	0	0	0	0	0	0	
0x48	SYSCFG_CFGR4	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ADC34_EXT14_RMP				ADC34_EXT11_RMP				ADC34_EXT15_RMP				
	Reset value																						0	0	0	0	0	0	0	0	0	0	0	0	

Refer to [Section 3.2 on page 53](#) for the register boundary addresses.